

2 Biodiversity can be assessed at three different levels. One of these is the genetic variation within each species.

(a) Outline **two** other levels at which biodiversity can be measured.

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To calculate the genetic variation that exists within a species, scientists:

- obtain DNA sequences from many individuals of one species
- count the number of nucleotides that differ when the sequences of two individuals are compared
- repeat this with different pairs of individuals.

This allows scientists to calculate the **mean** number of differences at every nucleotide position along the sequence (mean number of nucleotide differences per site).

(b) Explain why scientists use databases and computers to calculate the mean number of nucleotide differences per site.

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(c) Table 2.1 shows the mean number of nucleotide differences per site of some species.

Table 2.1

species	mean number of nucleotide differences per site
<i>Drosophila melanogaster</i> , fruit fly	0.0087
<i>Anopheles gambiae</i> , mosquito vector of malaria	0.0301
<i>Plasmodium falciparum</i> , malarial pathogen	0.0015
<i>Zea mays</i> , wild maize	0.0139

(i) State the **genus** name of the species that shows the most genetic variation.

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(ii) State how many kingdoms of organisms are represented in Table 2.1.

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(d) Genetic variation is considered important in the conservation of species. Low genetic variation is assumed to decrease the chance of the long-term survival of a species.

(i) Give reasons why low genetic variation may decrease the long-term survival of a species.

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Fig. 2.1 shows how the International Union for the Conservation of Nature (IUCN) categorises species according to their conservation status.

Common species with the lowest conservation status (least risk of extinction) are categorised as Least Concern (LC).

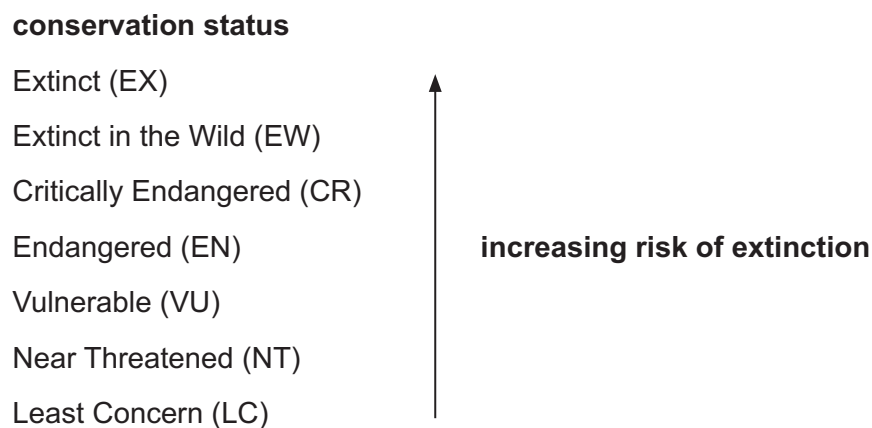


Fig. 2.1

(ii) Question 2(d) states that 'low genetic variation is assumed to decrease the chance of the long-term survival of a species'.

Predict the relationship between genetic variation and conservation status if this assumption is true.

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Fig. 2.2 shows the mean number of nucleotide differences per site of some species and sub-species of mammal and their conservation status.

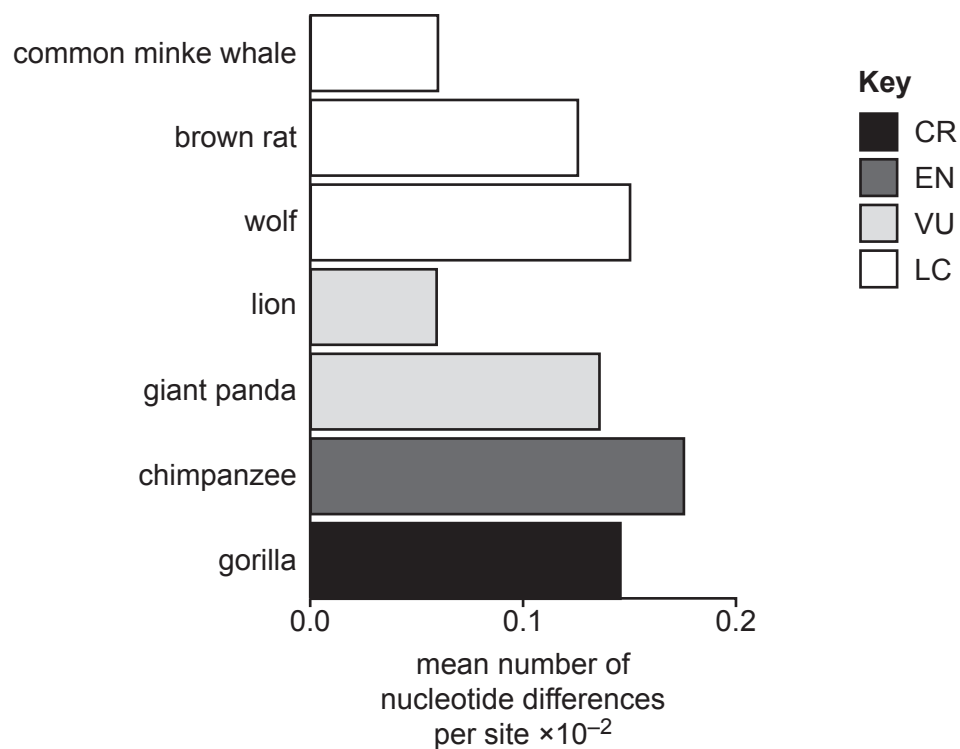


Fig. 2.2

(iii) Assess whether the data in Fig. 2.2 provide support for the prediction you made in 2(d)(ii).

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[Total: 14]